Maximum and minimum values

1 Consider the function $f(x) = 2x^3 - 12x^2 + 18x + 3$.

- **a** Find the x-coordinates of the turning points.
- **b** State the coordinates of the local maximum.
- **c** State the coordinates of the local minimum.
- **d** State the absolute maximum value on $[0, 7]$.
- **e** State the absolute minimum value on $[0, 7]$.

2 Consider the function $y = 5x^3 + 7x^2 + 3x + 4$ with restricted domain of $[-1, 1]$.

- **a** Find the y-intercept.
- **b** Find the stationary points.
- **c** Determine the nature of the stationary points.
- **d** Find any points of inflection and classify them.
- **e** Describe the behaviour of the function as $x \to \pm\infty$.
- **f** Find the maximum and minimum value of the function for the restricted domain.

3 Consider the function $y = (x^2 - 1)^2 + 4$ with restricted domain of $[-2, 2]$.

- **a** Find the y-intercept.
- **b** Find the x-intercept(s).
- **c** Determine whether the function is odd or even.
- **d** Find the stationary points.
- **e** Determine the nature of the stationary points.
- **f** Find any points of inflection and classify them.
- **g** Describe the behaviour of the function as $x \to \pm\infty$.
- **h** Find the maximum and minimum value of the function for the restricted domain.

4 Consider the function $y = 3x^4 + 4x^3 - 12x^2 - 1$.

- **a** Find the y-intercept.
- **b** Find the stationary points.
- **c** Determine the nature of the stationary points.
- **d** Find any points of inflection and classify them.
- **e** Describe the behaviour of the function as $x \to \pm\infty$.
- **f** Find the maximum and minimum value of the function for the restricted domain $[-1, 1]$.

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Graphs of functions

5 Consider the function $y = -x^2 + 8x - 21$ with restricted domain of $[0, 6]$.

- **a** Find the y-intercept.
- **b** Find the stationary points.
- **c** Determine the nature of the stationary points.
- **d** Find the maximum and minimum value of the function for the restricted domain.
- **e** Sketch the function on a number plane.

6 Consider the function $y = x^4 - 2x^2 + 1$ with restricted domain of $[-2, 3]$.

- **a** Find the y-intercept.
- **b** Find the x-intercept(s).
- **c** Determine whether the function is odd or even.
- **d** Find the stationary points.
- **e** Determine the nature of the stationary points.
- **f** Find any points of inflection and classify them.
- **g** Find the maximum and minimum value of the function for the restricted domain.
- **h** Sketch the function on a number plane.

7 Consider the function $y = (2x - 1)^2(1 - x)$ with restricted domain of $[0, 2]$.

- **a** Find the y-intercept.
- **b** Find the x-intercept(s).
- **c** Find the stationary points.
- **d** Determine the nature of the stationary points.
- **e** Find any points of inflection and classify them.
- **f** Describe the behaviour of the function as $x \to \pm\infty$.
- **g** Find the maximum and minimum value of the function for the restricted domain.
- **h** Sketch the function on a number plane.

8 Consider the function $y = 9x^2 + 18x - 16$.

- **a** Find the y-intercept.
- **b** Find the x-intercept(s).
- **c** Find the stationary points.
- **d** Determine the nature of the stationary points.
- **e** Sketch the function on a number plane.

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9 Consider the function $y = 8(x-1)^2(x+2)$.

- **a** Find the stationary points.
- **b** Determine the nature of the stationary points.
- **c** Determine the point of inflection.
- **d** Sketch the function on a number plane.

10 For each of the following functions:

- **i** Find any stationary points and determine their nature.
- **ii** Determine the points of inflection.
- **iii** Sketch the graph showing stationary points and points of inflection.
- **iv** Write the domain for which the function is decreasing.

a $f(x) = 8 - x^3$

b $f(x) = 8 - x - x^3$

c $f(x) = x^3 - 3x^2 - 9x + 11$

d $y = x^3 - 6x^2 - 3$

e $f(x) = (x^2 - 9)^2 + 3$

f $f(x) = x^3 - 5x^2 + 3x - 5$

11 For each of the following functions:

- **i** Find any stationary points and determine their nature.
- **ii** Determine the points of inflection.
- **iii** Sketch the graph showing stationary points and points of inflection.
- **iv** Write the domain for which $\frac{dy}{dx}$ is positive.

a $f(x) = x^4(3x - 10)$

b $f(x) = x^2(x + 3)$

12 For each of the following functions:

- **i** Find any stationary points and determine their nature.
- **ii** Determine the points of inflection.
- **iii** Sketch the graph showing stationary points and points of inflection.
- **iv** Write the domain over which the curve is concave up and concave down.

a $f(x) = 3x^2 - x^3 - 4$

b $f(x) = x^3 - 3x^2 - 9x + 11$

c $f(x) = x^3(x - 4)$

d $f(x) = x^2(x^2 - 6x + 12)$

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13 Consider the function $y = 2(x+5)^3 + 3$.

- **a** Find the y-intercept.
- **b** Find the x-intercept(s).
- **c** Find the stationary points.
- **d** Determine the nature of the stationary points.
- **e** Find any points of inflection and classify them.
- **f** Describe the behaviour of the function as $x \to \pm\infty$.
- **g** Sketch the function on a number plane.

14 Consider the function $y = 2 + 9x - 3x^2 - x^3$.

- **a** Find the y-intercept.
- **b** Find the stationary points.
- **c** Determine the nature of the stationary points.
- **d** Find any points of inflection and classify them.
- **e** Describe the behaviour of the function as $x \to \pm\infty$.
- **f** Sketch the function on a number plane.

15 Consider the function $y = -8(x-1)^2(x+2)$.

- **a** Find the y-intercept.
- **b** Find the x-intercept(s).
- **c** Find the stationary points.
- **d** Determine the nature of the stationary points.
- **e** Find any points of inflection and classify them.
- **f** Describe the behaviour of the function as $x \to \pm\infty$.
- **g** Sketch the function on a number plane.

16 Consider the function $y = (2x+1)(x+2)^4$.

- **a** Find the y-intercept.
- **b** Find the x-intercept(s).
- **c** Find the stationary points.
- **d** Determine the nature of the stationary points.
- **e** Describe the behaviour of the function as $x \to \pm\infty$.
- **f** Sketch the function on a number plane.

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17 Consider the function $y = \dfrac{8}{x^2 + 4}$.

a Find the y-intercept.

b Find the x-intercept(s).

c Determine whether the function is odd or even.

d Find the stationary points.

e Determine the nature of the stationary points.

f Describe the behaviour of the function as $x \to \pm\infty$.

g Sketch the function on a number plane.

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